

Which Regression Line Fits?

Unit 5 · Topic 5.5 · THE PROBLEM

Suppose a solar-farm analyst uses sunshine hours x to predict daily generation y in megawatt-hours (MWh). The table summarizes 12 observed days.

Quinn: “Using $b = r(s_y/s_x)$ gives $\hat{y} = 5.6 + 3.2x$. It passes through (7, 28), so it must be the LSRL. It explains 80 percent of the variation.”

Rowan: “Using $b = r(s_x/s_y)$ gives $\hat{y} = 26.6 + 0.2x$. It also passes through (7, 28), so it is equally good. Explained variation is 64 percent.”

Solar-farm summary (12 observed days)

x range	$\bar{x}$	$\bar{y}$	s_x	s_y	r
4–10 h	7 h	28 MWh	1.5 h	6 MWh	0.80

FIRST TAKE — one sentence before you work the parts

The two lines agree at 7 sunshine hours. Does that make them equally useful at other times? Give your first thought in one sentence.

- DOK 1** (a) State the mean point that every least-squares regression line must contain.
- DOK 2** (b) Calculate the regression slope and intercept, and r^2 . Give the slope’s units.
- DOK 3** (c) Which parts of Quinn’s and Rowan’s claims should a report keep? Use the slope formula to choose a line, explain why their shared mean point does not make both lines correct, and interpret the correct percent of variation explained. Write 3–4 sentences.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.